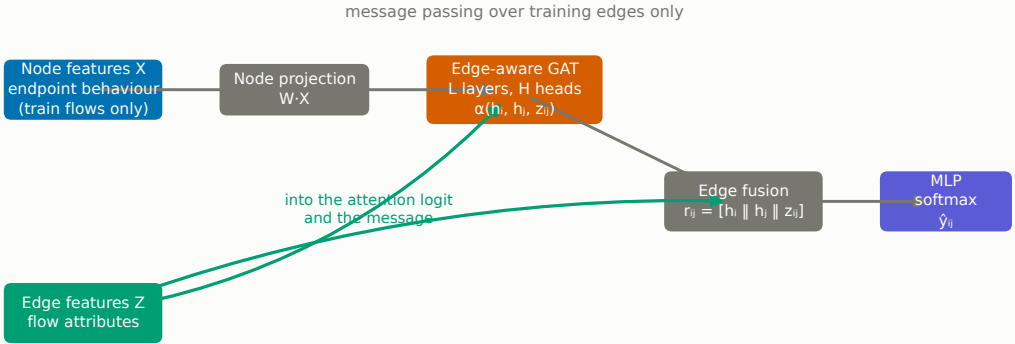
E-GAT-IDS: edge-level classification on the communication graph
message passing over training edges only
Node features X
endpoint behaviour
(train flows only)
Node projection
W·X
Edge-aware GAT
L layers, H heads
α(hᵢ, hⱼ, zᵢⱼ)
Edge fusion
rᵢⱼ = [hᵢ ‖ hⱼ ‖ zᵢⱼ]
MLP
softmax
ŷᵢⱼ
into the attention logit
and the message
Edge features Z
flow attributes